

$$\begin{aligned}
D &= \sum_{i=1}^N \sum_{k=1}^k r_{ik} ||x^{(i)} - u_k||^2 \\
\rightarrow \frac{\delta D}{\delta u_k} &= \sum_{i=1}^N r_{ik} \frac{\delta}{\delta u_k} ||x^{(i)} - u_k||^2 \\
||x - u||^2 &= x^T x - 2x^T u + u^T u \\
\frac{\delta}{\delta u} ||x - u||^2 &= -2x + 2u \\
&\rightarrow -2(x - u) \\
\rightarrow \frac{\delta D}{\delta u_k} &= \sum_{i=1}^N -2r_{ik}(x^{(i)} - u_k) \\
\rightarrow \frac{\delta D}{\delta u_k} &= -2 \sum_{i=1}^N r_{ik}(x^{(i)} - u_k)
\end{aligned}$$